

		All in USD			
	Count	Cost Per Unit	Total Cost (before taxes & shipping)	Name + Link	Notes
ELECTRONICS	10	12.1	121	TowerPro Servo Motor MG996R 180	
	1	110	110	Raspberry Pi 5 (4GB Ram)	
	4	20	80	D2822 Fixed Wing Motor	
	2	24.99	49.98	Power Distribution Board 2 (8 connectors)	
	4	12.02	48.08	Bidirectional Brushless ESC	
	1	35	35	Hosyond 5 Inch Touchscreen IPS MIPI DSI Display	
	1	30	30	Turnigy 5000mAh 3S 30C 11.1V Lipo Battery Pack w/ XT90	Any 11.1V Lipo battery will work. mAh is just a measurement of how long the battery will last. Higher mAh is equal to longer battery life.
	1	16.64	16.64	5MP 1080P Optional Night Vision Wide Angle IR-CUT FF AF Camera	
	2	12	24	Power Distribution Board 1 (6 connectors)	
	1	10	10	2pcs BEC UBEC 5V 5A	You need 2 UBECs. This offering provides both.
	4	2.5	10	TowerPro MG90S Servo Motor	
	1	9.36	9.36	6mm Precision Mini Micro Planetary Reduction Reducer Gear Motor	
	1	74	74	PS5 Controller	You can choose another controller if you'd like, if it doesn't work upon initialization just ensure you modify the code to make it compatible.
	1	5	5	PCA9685PW	Generic PCA9685 brand. Any other model will work too (including the Adafruit model).
	1	3	3	MPU-6050	
MATERIALS	1	45.51	45.51	SEBS Filament for wheel sleeves	
	1	19.99	19.99	White PLA Basic Filament (1kg) Bambu Lab US Store	
	1	19.99	19.99	Grey PLA Basic (1kg) Bambu Lab US Store	
	1	17.72	19.99	Nylon Filament (500g)	
	1	7.1	7.1	White Lithium Grease	Lube for contact points. Use on spinning axles and metal to metal contact against bearings.
	1	7	7	Two part epoxy	This is used everywhere a bearing is located to retain them. Let the epoxy cure before budging the bearings.

	1	5.8	5.8	Loctite 242 for screws	This is used everywhere a screw or nut is located to prevent loosening against vibrations. Let the loctite cure before budging the screw or nut.
	1	4.22	4.22	Tapping fluid	
HARDWARE	2	4.53	9.06	83T HTD 3M Belts 6mm Width	
	4	3.73	14.92	60T HTD 3M Belts 6mm Width	
	4	3.51	14.04	55T HTD 3M Belts 6mm Width	
	1	1.05	1.05	Wires (10 awg)	
	5	0.8	4	Wires (12 awg)	
	2	1.21	2.42	Wires (18 awg)	
	8	1.35	10.8	M4 heat set inserts	The exact model doesn't matter, just ensure the height exceeds no more than 4.7mm.
	1	12.69	12.69	4mm ID x 8mm OD x 250mm shaft	You will need to be able to cut the shaft accurately into smaller sections.
	18	0.6	10.8	8mm ID x 16 OD x 5mm WD (f688z Flanged Bearings)	
	1	7.75	7.75	Bungee Cord	
	1	5.33	5.33	Velcro	
	1	5.24	5.24	Scotch Extreme Mounting Tape	
	1	2.89	2.89	M2.5 washer	
	12	0.3158333333	3.79	10-32 washers	
	1	1.48	1.48	Electrical Tape	
	1	1.77	1.77	Shrink Tubing	
	1	5.05	5.05	PTFE thrust bearing slider	Must be identical to linked one.
	5	5	25	Sets of M2, M2.5, M3, M4, 10-32 screws	(You can get a set of both rounded & socket head screws online or match the screws you buy by length, size, and amount to the CAD 1:1 by searching "screw" and in the assembly and counting.)
	4	8.37	33.48	128-2Z Bearing	
TOOLS	1	1	1	10-32 tap (for tapping shafts)	
	1	1	1	10-32 tap handle	
	1	10.99	10.99	Soldering Iron & Solder Kit	
	1	6.77	6.77	Wrench Set	
	1	15.09	15.09	Allen Key Set	
	1	20.41	20.41	Wire Stripper, Cutter, and Crimper tool	You will use this for ensuring proper wire connections.

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